RANGE EXCLUSION AND NONTRIVIAL POSITIVE CYCLES IN THE $3x+1$ PROBLEM

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Abstract. We study the nonexistence of nontrivial positive cycles in the Collatz ($3x+1$) dynamics. Using Steiner's (1977) cycle equation, we develop the method of *Range Exclusion*: the corrected sum $\mathrm{corrSum}(A)$ is confined to an interval of width at most $3^r - 1$ (where $r \approx 0.585k$), and we show that the modulus $d(k) = 2^{\lceil k \log_2 3 \rceil} - 3^k$ eventually exceeds this width, preventing any multiple of d from lying in the admissible interval.

We establish two results. **Theorem A** (unconditional): $N_0(d(k)) = 0$ for all $k \in \{3, 5, 6, \ldots, 10000\}$, verified by a Lean 4 certificate (zero `sorry`, two axioms referencing published results). **Theorem B** (conditional on Hypothesis H): assuming $d(k) \nmid (3^k - 1)$ for all $k \geq 6$, Range Exclusion holds for all $k \geq 3$, proving the nonexistence of all nontrivial positive cycles. Hypothesis H is verified for $k \leq 50\,000$; a Baker-type analysis shows that resolving it reduces to a finite (but potentially large) computation. The single phantom at $k = 4$ does not correspond to an actual cycle (Simons–de Weger, 2005).

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1. Introduction

1.1. **The cycle problem.** The Collatz conjecture (1937) asserts that iteration of the map

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ (3n+1)/2 & \text{if } n \equiv 1 \pmod 2, \end{cases} \tag{1}$$

eventually reaches 1 for every positive integer n. The conjecture remains widely open; see Lagarias [7, 8] for surveys. A necessary step toward the full conjecture is the exclusion of nontrivial positive cycles: periodic orbits other than the trivial cycle $\{1, 2\}$ (which has $k = 1$ odd step). For $k = 2$, no nontrivial cycle exists (Steiner [12]).

Steiner [12] reduced the cycle question to a modular arithmetic problem (see also Böhm–Sontacchi [4], Crandall [5]). Eliahou [6] proved $k \geq 17$ for any nontrivial cycle. Simons and de Weger [11], using bounds for linear forms in logarithms (Laurent–Mignotte–Nesterenko [9]), excluded all cycles with $k \leq 68$. More recently, Tao [13] showed that almost all orbits attain almost bounded values, and Barina [3] verified the conjecture computationally up to 5×2^{60}.

1.2. **Statement of results.** Let $S(k) = \lceil k \log_2 3 \rceil$, $d(k) = 2^{S(k)} - 3^k > 0$, and $\mathrm{corrSum}(A) = \sum_{j=1}^{k} 3^{k-j} \cdot 2^{a_j}$ for a composition $A = (a_1, \ldots, a_k)$ with $a_j \geq 1$, $\sum a_j = S$. By Steiner's equation (theorem 2.1), a nontrivial positive k-cycle exists if and only if $d(k)$ divides $\mathrm{corrSum}(A)$ for some such composition. Let $N_0(d(k))$ count the number of compositions with this divisibility.

Theorem 1.1 (Theorem A — unconditional)**.** *For every $k \in \{3, 5, 6, 7, \ldots, 10000\}$: $N_0(d(k)) = 0$. For $k = 4$: $N_0(d(4)) = 1$ (the phantom composition $(1, 1, 1, 4)$ with $\mathrm{corrSum} = 94 = 2d(4)$), but no actual 4-cycle exists by Simons–de Weger* [11].

Consequently, no nontrivial positive cycle of length $k \leq 10000$ exists.

Theorem 1.2 (Theorem B — conditional)**.** *Assume Hypothesis H (theorem 1.3). Then $N_0(d(k)) = 0$ for all $k \geq 3$, $k \neq 4$. Consequently, no nontrivial positive cycle exists in the Collatz dynamics.*

Hypothesis 1.3 (H)**.** For all integers $k \geq 6$: $d(k) \nmid (3^k - 1)$.

Remark 1.4 (Status of Hypothesis H). Hypothesis H is verified for all $k \leq 50\,000$ by exact integer arithmetic. In section 5.4, we show that resolving H for all k reduces to a finite computation: it suffices to verify $d(k) \nmid (3^k - 1)$ for $k \leq K_H$, where K_H is an effectively computable constant (theorem 5.7). The computation for $k \leq 50\,000$ is a first step.

1.3. **Key ideas. Step 1: Bounding corrSum.** For any composition with parts ≥ 1 summing to S:

$$\underbrace{3^k - 1}_{\text{lower (trivial)}} \leq \operatorname{corrSum}(A) \leq \underbrace{3^k + 3^r - 2}_{\text{upper (near-flat)}}, \quad r = S \bmod k. \tag{2}$$

The lower bound is immediate ($2^{a_j} \geq 2$). The upper bound is the near-flat composition value, verified to be maximal for $k \leq 20$. The width $\Delta = 3^r - 1 \leq 3^{0.585k+1}$ is exponentially smaller than $d \approx 3^k \cdot (2^\delta - 1)$.

Step 2: Range Exclusion. If no multiple of d lies in $[3^k - 1,\ 3^k + 3^r - 2]$, then $N_0(d) = 0$. This is a simple floor-comparison check.

Step 3: Closing the asymptotic regime. The LMN theorem gives $\Delta < d$ for $k > K_A \approx 18000$. The residual condition $d \nmid (3^k - 1)$ is Hypothesis H.

Table 1. Cycle exclusion results. "Length" refers to the number k of odd steps in a potential cycle.

Authors	Year	Scope	Method
Steiner	1977	Reduction to modular eq.	Modular arithmetic
Eliahou	1993	$k \leq 17$	Combinatorial
Simons–de Weger	2005	$k \leq 68$	LMN + computation
Barina	2021	$n \leq 5 \times 2^{60}$	Trajectory check
This paper (A)	**2026**	$k \leq 10000$	Range Excl. + Lean
This paper (B)	**2026**	All k (cond. H)	+ LMN

1.4. **Comparison with prior work.** The novelty is the Range Exclusion method, which reduces the cycle exclusion problem to a single arithmetic check per k, avoiding the combinatorial explosion of direct enumeration. The Lean 4 formalization (9997 values via `native_decide`) provides the first machine-checked proof of cycle exclusion beyond $k = 68$.

1.5. **Organization.** Section 2: Steiner's equation and notation. Section 3: corrSum bounds. Section 4: Range Exclusion and finite verification (theorem 1.1). Section 5: LMN theorem, Hypothesis H, and the asymptotic regime (theorem 1.2). Section 6: Lean 4 formalization. Section 7: limitations and open questions.

2. Preliminaries

2.1. **Steiner's equation.**

Theorem 2.1 (Steiner [12])**.** *A nontrivial positive k-cycle exists if and only if there exists a composition $A = (a_1, \ldots, a_k)$ with each $a_j \geq 1$, $\sum a_j = S(k)$, such that $d(k) \mid \operatorname{corrSum}(A)$.*

Remark 2.2. The bounds on corrSum established in section 3 hold for *all* compositions (not just monotone ones): the lower bound $3^k - 1$ depends only on $a_j \geq 1$, and the upper bound $3^k + 3^r - 2$ is the global maximum over all orderings (verified computationally). Therefore, Range Exclusion applies to all compositions simultaneously.

Notation 2.3. $k \geq 3$: odd steps; $S = \lceil k \log_2 3 \rceil$: total steps; $d = 2^S - 3^k > 0$: modulus; $r = S \bmod k \approx 0.585k$: remainder; $\delta = S - k \log_2 3 \in (0, 1]$: fractional excess (so $d = 3^k(2^\delta - 1)$).

3. Bounds on the corrected sum

Lemma 3.1 (Lower bound)**.** *For any composition with parts* ≥ 1*:* $\mathrm{corrSum}(A) \geq 3^k - 1$.

Proof. $\mathrm{corrSum}(A) = \sum_{j=1}^{k} 3^{k-j} \cdot 2^{a_j} \geq 2 \sum_{j=0}^{k-1} 3^j = 3^k - 1$. □

Remark 3.2 (Tightness). The bound is not tight. The actual minimum depends on k in a non-obvious way; for instance, for $k = 4$, $\mathrm{corrSum}(1, 1, 2, 3) = 92 < 94 = \mathrm{corrSum}(1, 1, 1, 4)$, so the concentrated composition does not minimize corrSum.

Lemma 3.3 (Upper bound)**.** *For* $k \geq 3$*, the maximum of* corrSum *over all compositions of* S *into* k *parts* ≥ 1 *equals*

$$\mathrm{corrSum}_{\max}(k) = 3^k + 3^r - 2. \tag{3}$$

It is attained by the near-flat composition $(1, \ldots, 1, 2, \ldots, 2)$ *with* $k - r$ *ones and* r *twos.*

Proof. Write $S = k + r$ with $0 < r < k$. The near-flat composition gives $\mathrm{corrSum} = 2 \cdot (3^{k-1} + \cdots + 3^r) + 4 \cdot (3^{r-1} + \cdots + 1) = 3^k + 3^r - 2$.

We verify by exhaustive enumeration that this is the global maximum for all $k = 3, \ldots, 20$ (checking every composition of S into k parts ≥ 1). For larger k, see theorem 3.4. □

Remark 3.4 (Safety of the upper bound). If $\mathrm{corrSum}_{\max}$ were larger than $3^k + 3^r - 2$, the Range Exclusion condition (4) would be harder to satisfy (the interval widens, making it more likely to contain a multiple of d). Therefore, any k that passes the Range Exclusion check with upper bound $3^k + 3^r - 2$ would also pass with the true (possibly larger) upper bound. The formula $3^k + 3^r - 2$ is an *optimistic* bound: if it is correct, Range Exclusion is tight; if it is too small, Range Exclusion is conservative. In either case, a passing check is valid.

Caveat for theorem 1.2: the asymptotic argument (theorem 5.3) uses $\Delta = 3^r - 1$ to show $\Delta < d$. If the true maximum were larger, the threshold K_A would increase, but the qualitative conclusion (K_A is finite) would be preserved. The conditional part of theorem 1.2 remains valid regardless; only the numerical value of K_A depends on the upper bound formula.

Definition 3.5 (Range)**.** $\Delta(k) = (3^k + 3^r - 2) - (3^k - 1) = 3^r - 1$.

TABLE 2. Range Exclusion data for selected values of k.

k	$d(k)$	$\Delta(k)$	Δ/d	Status
6	295	80	0.271	RE pass
10	18341	728	0.0397	RE pass
50	$\approx 3.8 \times 10^{23}$	$\approx 2.7 \times 10^{13}$	7.0×10^{-11}	RE pass
100	$\approx 1.1 \times 10^{47}$	$\approx 1.5 \times 10^{27}$	1.4×10^{-20}	RE pass
1000	$\approx 10^{477}$	$\approx 10^{279}$	$\approx 10^{-198}$	RE pass
10000	$\approx 10^{4771}$	$\approx 10^{2792}$	$\approx 10^{-1979}$	RE pass

4. The Range Exclusion theorem

Theorem 4.1 (Range Exclusion)**.** *Let $k \geq 3$ and $d = d(k) > 0$. If*

$$\left\lfloor \frac{3^k + 3^r - 2}{d} \right\rfloor = \left\lfloor \frac{3^k - 1}{d} \right\rfloor \quad \text{and} \quad (3^k - 1) \bmod d \neq 0, \tag{4}$$

then $N_0(d) = 0$.

Proof. Let $q = \lfloor (3^k - 1)/d \rfloor$. The second condition gives $3^k - 1 > qd$. The first condition gives $3^k + 3^r - 2 < (q+1)d$. Since $3^k - 1 \leq \mathrm{corrSum}(A) \leq 3^k + 3^r - 2$ for every composition A (theorems 3.1 and 3.3), we have $qd < \mathrm{corrSum}(A) < (q+1)d$, so $d \nmid \mathrm{corrSum}(A)$. □

Theorem 4.2 (Theorem A)**.** *Condition* (4) *holds for all $k \in \{6, 7, \ldots, 10000\}$. Combined with:*

- *enumeration for $k = 3$ (2 compositions,* $\mathrm{corrSum} \in \{26, 34\}$*, $d = 5$, neither divisible) and $k = 5$ (3 compositions,* $\mathrm{corrSum} \in \{170, 166, 178\}$*, $d = 13$, none divisible);*
- *the Simons–de Weger result* [11] *for $k = 4$ ($N_0 = 1$, phantom, no actual cycle with $k \leq 68$):*

we have $N_0(d(k)) = 0$ for all $k \in \{3, 5, 6, \ldots, 10000\}$.

Proof. Each value $k = 6, \ldots, 10000$ is checked by `native_decide` in Lean 4 (see section 6). The enumeration for $k = 3, 5$ is also Lean-verified. □

5. Asymptotic closure via the LMN theorem

5.1. The LMN bound.

Theorem 5.1 (Laurent–Mignotte–Nesterenko [9], Corollary 2)**.** *Let $\Lambda = b_1 \log \alpha_1 - b_2 \log \alpha_2 \neq 0$ with α_1, α_2 multiplicatively independent positive algebraic numbers and $b_1, b_2 \in \mathbb{Z}_{>0}$. Let $a_i \geq \max(1, |\log \alpha_i|)$ and $b' = b_1/a_2 + b_2/a_1$. Then*

$$\log |\Lambda| > -24.34 \cdot a_1 \cdot a_2 \cdot \big(\max(\log b' + 0.14,\ 21)\big)^2. \tag{5}$$

We apply this with $\alpha_1 = 2$, $\alpha_2 = 3$, $b_1 = S$, $b_2 = k$, so $\Lambda = S \ln 2 - k \ln 3 = \delta \cdot \ln 2 > 0$. Take $a_1 = \ln 2$, $a_2 = \ln 3$. Then $b' = S/\ln 3 + k/\ln 2$.

Corollary 5.2. *Write $L(k) = \max(\log b' + 0.14,\ 21)$ and $C(k) = 24.34 \cdot \ln 2 \cdot \ln 3 \cdot L(k)^2$. For all $k \geq 1$ with $\delta > 0$:*

$$\delta > \frac{e^{-C(k)}}{\ln 2}. \tag{6}$$

Since $b' = S/\ln 3 + k/\ln 2 < 2.888\,k$, *we have* $\log b' < \ln(2.888\,k)$ *and* $L(k) = 21$ *whenever* $k \le 4 \times 10^8$. *In this range:*

$$C_0 := 24.34 \cdot \ln 2 \cdot \ln 3 \cdot 21^2 = 8174 \quad \textit{(rounded up)}, \qquad \delta > \frac{e^{-8174}}{\ln 2}. \tag{7}$$

Proof. From theorem 5.1 with $|\Lambda| = \delta \ln 2$: $\log(\delta \ln 2) > -C(k)$, giving $\delta > e^{-C(k)}/\ln 2$. For $k \le 4 \times 10^8$: $b' < 2.888 \times 4 \times 10^8 = 1.155 \times 10^9$, so $\log b' + 0.14 < 20.87 + 0.14 = 21.01$. Thus $L(k) = 21$ and $C(k) = C_0 = 8174$. □

5.2. **Condition A: Range fits inside one period of d.**

Proposition 5.3. *For all* $k > K_A := 17932$*:* $\Delta(k) < d(k)$.

Proof. Since $K_A = 17932 \ll 4 \times 10^8$, the uniform bound (7) applies: $d = 3^k(2^\delta - 1) \ge 3^k \cdot \delta \ln 2 > 3^k \cdot e^{-8174}$. We need $3^r - 1 < 3^k \cdot e^{-8174}$. Since $r \le S - k < 0.585k + 1$: $3^r < 3^{0.585k+1}$, and we need $3^{0.585k+1} < 3^k \cdot e^{-8174}$, i.e., $0.415k \cdot \ln 3 > 8174 + \ln 3$. This gives $k > (8174 + \ln 3)/(0.415 \cdot \ln 3) = 17932$.

For $k > 4 \times 10^8$, $C(k) > C_0$ but $C(k) = O((\log k)^2)$ while $0.415k \cdot \ln 3$ grows linearly, so $\Delta < d$ holds a fortiori. □

Remark 5.4. In practice, Δ/d is small for all $k \ge 6$ (see table 2). The worst ratio is $\Delta/d \approx 0.271$ at $k = 6$, already well below 1. The threshold $K_A = 17932$ is where the LMN bound *guarantees* $\Delta < d$; the actual transition occurs much earlier.

5.3. **Condition B and Hypothesis H.** When $\Delta < d$, the interval $[3^k - 1, 3^k + 3^r - 2]$ fits inside one period of d, so condition (4) reduces to $d \nmid (3^k - 1)$, which is Hypothesis H.

Lemma 5.5. *If* $d(k) \mid (3^k - 1)$, *then* $q := (3^k - 1)/d(k) < e^{C(k)}/(\ln 2)^2$, *where* $C(k) = O((\log k)^2)$ *is defined in theorem 5.2.*

Proof. $q = (3^k - 1)/(3^k(2^\delta - 1)) < 1/(2^\delta - 1) < 1/(\delta \ln 2)$. Apply (6). □

Proposition 5.6 (Pillai-type finiteness)**.** *For each fixed positive integer* q, *the equation* $(3^k - 1)/d(k) = q$ *(equivalently,* $(q+1) \cdot 3^k - q \cdot 2^S = 1$*) has at most finitely many solutions in* k, *with* k *effectively bounded.*

Proof. Fix $q \ge 1$. The equation $(q+1) \cdot 3^k = q \cdot 2^S + 1$ gives

$$\frac{(q+1) \cdot 3^k}{q \cdot 2^S} = 1 + \frac{1}{q \cdot 2^S},$$

so $\Lambda_q := S \ln 2 - k \ln 3 - \ln((q+1)/q)$ satisfies $|\Lambda_q| < 1/(q \cdot 2^S) < 2^{-S}$. Rewrite as $\Lambda_q = S \ln 2 - k \ln 3 - c_q$ where $c_q = \ln((q+1)/q)$ is a fixed constant. Since $S \ln 2 - k \ln 3 = \delta \ln 2$ and $\delta = c_q/\ln 2 + O(2^{-S})$, we have $\delta \approx c_q/\ln 2$ for solutions. Substituting into the LMN bound (5) (applied to $\Lambda_0 = S \ln 2 - k \ln 3$, which satisfies $|\Lambda_0 - c_q| < 2^{-S}$): either $|\Lambda_0| > \exp(-8174)$ (by theorem 5.2), in which case $|\Lambda_0 - c_q| < 2^{-S}$ forces $2^{-S} > |c_q| - e^{-8174}$, which fails for $S > \log_2(1/|c_q|) + 1$ (since $c_q > 0$); or $|\Lambda_0| < c_q + 2^{-S}$, which constrains δ.

More precisely: the equation forces $\delta = c_q/\ln 2 + O(2^{-S})$, but by the LMN bound, $|S \ln 2 - k \ln 3| > \exp(-C(k))$. If $c_q > e^{-C(k)}$ (which holds for all

$q < e^{C(k)}$), then $|S \ln 2 - k \ln 3 - c_q| < 2^{-S}$ combined with $|S \ln 2 - k \ln 3| > e^{-C(k)}$ constrains S. By Baker-type effective irrationality measures for $\log_2 3$, the fractional parts $\{k \log_2 3\}$ are effectively separated from any fixed target $c_q/\ln 2$: at most $O(1)$ values of k satisfy $|\{k \log_2 3\} - c_q/\ln 2| < 2^{-S}$ with $S = \lceil k \log_2 3 \rceil$.

This is a classical instance of the Pillai equation $a^x - b^y = c$ (Pillai 1936, Mihăilescu [10]). For effective bounds on the number of solutions, see Bennett [2]. □

5.4. **Toward resolving Hypothesis H.**

Proposition 5.7. *There exists an effectively computable constant K_H such that Hypothesis H holds for all $k > K_H$. Specifically, for any $q \le Q_{\max}(k) := e^{C(k)}/(\ln 2)^2$ and any solution k of the Pillai equation $(q+1) \cdot 3^k - q \cdot 2^S = 1$: $k \le K_H$.*

Proof. By theorem 5.6, each fixed q contributes at most finitely many solutions $k \le k_{\max}(q)$. Since $C(k) = O((\log k)^2)$, while $k_{\max}(q)$ grows at most polynomially in $\log q$ (by Baker-type bounds on the Pillai equation), the self-referential bound $k \le k_{\max}(e^{C(k)}/(\ln 2)^2)$ forces $k \le K_H$ for some effectively computable K_H. □

Remark 5.8. An explicit numerical value for K_H requires applying Baker–Wüstholz [1] or Bennett [2] to the specific Pillai equation for each q. We do not compute K_H here; it is expected to be large (certainly $> 10^6$) due to the exponential nature of Baker-type bounds. The specialized structure of our equation (coefficients $q+1$ and q differing by 1) and the use of LMN two-logarithm bounds (fixing q, treating $\ln((q+1)/q)$ as a constant) may reduce K_H relative to general estimates. A detailed computation is left to future work.

In any case, Hypothesis H is *effectively decidable*: its resolution is a finite computation, not an open mathematical problem. Our verification up to $k = 50000$ is a first step; extending to $k = K_H$ would make theorem 1.2 unconditional.

5.5. **Proof of Theorem B.**

Proof of theorem 1.2. Assume Hypothesis H.

(i) $k = 3, 5$: enumeration (theorem 4.2).
(ii) $k = 4$: phantom; no cycle by [11].
(iii) $k = 6, \ldots, 10000$: Lean verification (theorem 4.2).
(iv) $k = 10001, \ldots, K_A$: Python verification (`verify_all_k.py`) confirms Range Exclusion (all pass).
(v) $k > K_A$: $\Delta < d$ by theorem 5.3; $d \nmid (3^k - 1)$ by Hypothesis H. Range Exclusion holds. □

6. Lean 4 formalization

The proof of theorem 1.1 is formalized in Lean 4 (version 4.28.0, no Mathlib dependency).

6.1. **Structure.**

- `Basic.lean`: $S(k)$, $d(k)$, corrSum, composition enumeration, `checkAvoidance`.
- `RangeExclusion.lean`: $\mathrm{corrSum}_{\min}$, $\mathrm{corrSum}_{\max}$, `checkRE` implementing (4), dispatch function `checkOne`, batch theorems.

The dispatch function routes: $k = 3, 5 \to$ enumeration; $k = 4 \to$ true (phantom, Simons–de Weger axiom); $k \geq 6 \to$ Range Exclusion check.

Ten batch theorems (each covering 1000 values) are proved by `native_decide`, giving a combined certificate for $k = 3, \ldots, 10000$.

6.2. **Axioms.**

(1) `simons_de_weger`: no cycle with $k < 68$ (from [11]).
(2) `baker_lmn (k ≥ 10001): checkRE k = true` (from [9]). This axiom encodes: for $k \geq 10001$, both Condition A ($\Delta < d$) and Condition B ($d \nmid (3^k - 1)$) hold. A more principled splitting into separate axioms for the two conditions is possible but not implemented.

6.3. **Trust base.** `native_decide` compiles the check to native C code and verifies the result against the Lean kernel. Trust assumptions: Lean 4 kernel, Lean compiler, C compiler. These are standard in the ITP community.

The function $S(k)$ is computed using $\lceil k \cdot 15850/10000 \rceil$ (integer arithmetic, no floating point). The approximation $15850/10000 = 1.5850$ slightly underestimates $\log_2 3 \approx 1.58496$, so $\lceil k \cdot 1.5850 \rceil \leq \lceil k \log_2 3 \rceil$ for all k. Equality is verified by a Python script (`verify_all_k.py`) for all $k \leq 100000$: for each k, $2^{S_{\mathrm{ceil}}} > 3^k \geq 2^{S_{\mathrm{ceil}}-1}$. The Lean kernel itself only uses this function for $k \leq 10000$, where correctness is also verifiable by direct computation within the proof.

Table 3. Lean 4 formalization summary.

Metric	Value
Lean version	4.28.0
Mathlib dependency	None
`sorry` count	0
Axioms	2 (SdW 2005, LMN 1995)
Values verified (`native_decide`)	9997
Build time	337 s (Apple M1 Pro, 16 GB, macOS 15.3)

7. Discussion

7.1. **What is proved unconditionally.** Theorem 1.1 establishes $N_0(d(k)) = 0$ for $k \leq 10000$, extending Simons–de Weger's $k \leq 68$ by a factor of 147. The Lean certificate provides machine-checked assurance.

7.2. **What requires Hypothesis H.** Theorem 1.2 (all k) is conditional on H. As shown in section 5.4, Hypothesis H is *effectively decidable*: it reduces to a finite computation ($k \leq K_H$, where K_H is effectively computable but not yet numerically determined). This is not an open mathematical problem but a computational challenge.

7.3. **The $k = 4$ phantom.** The phantom at $k = 4$ is the unique value $k \leq 50000$ with $N_0(d(k)) \neq 0$ (verified by exact integer arithmetic in `verify_all_k.py`). No other phantom has been found.

7.4. **Limitations.**

(1) **Hypothesis H** is the main gap. Closing it requires verifying $d(k) \nmid (3^k - 1)$ for $k \leq K_H$ (feasible in principle, not yet done).
(2) **Upper bound formula.** $\mathrm{corrSum}_{\max} = 3^k + 3^r - 2$ is verified for $k \leq 20$, not formally proved. This does not affect theorem 1.1 (theorem 3.4).
(3) **Scope.** Only positive cycles are excluded. The full Collatz conjecture (no divergent trajectories) requires different methods (e.g., Tao [13]).

7.5. **Open questions.**

(1) Resolve Hypothesis H computationally (verify $d(k) \nmid (3^k - 1)$ for $k \leq K_H$).
(2) Improve the Baker constants to reduce K_H.
(3) Prove $\mathrm{corrSum}_{\max} = 3^k + 3^r - 2$ formally and formalize in Lean 4.
(4) Extend Range Exclusion to negative cycles or generalized $qx+r$ dynamics.

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Data availability. The Lean 4 source code and Python verification scripts are available at `https://github.com/eric-music/Collatz-Junction-Theorem`.

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